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National Institute of Technology Goa

Programme Name: B.Tech.-I Year

Mid Semester Examinations, March 2025

Course Name: **Differential Equations and Vector Calculus**

Date: 03/03/2025

Duration: 1 Hour 30 Minutes

Course Code: **MA150**

Time: 10.00 AM - 11.30 AM

Max. Marks: 50

ANSWER ALL QUESTIONS

Part A (Ordinary Differential Equations)

- (a) Solve the equation $t^2 y' = y - t$ with the condition $y(0) = 0$. Does it have an analytic solution?
(b) Can you replace the equation $\frac{dy}{dt} = f\left(\frac{y}{t}\right)$ into equivalent separable form.
(c) Find the general solution of the equation $\frac{dy}{dt} = 2\left(\frac{y}{t}\right) + \left(\frac{y}{t}\right)^2$. [5]
- Solve the differential equation $(8x^2y^3 - 2y^4)dx + (5x^3y^2 - 8xy^3)dy = 0$ by first finding integrating factor. [5]
- Let $y = y_c + y_p$ is the solution of $(D^4 + 2D^2 + 1)y = x^2 \cos x$. then find y_c and y_p . [5]
- Solve the differential equation $y'' + 2y' + y = e^{-x} \log x$. [5]
- For the initial value problem $\frac{dy}{dx} = \frac{3x-6y+2}{x-2y+1}$, $y(0) = 0$, determine the interval of existence of its solution given that $R : |x| \leq \frac{1}{4}, |y| \leq \frac{1}{4}$. [5]

Part B (Multiple Integrals)

- Evaluate $\int_0^{4a} \int_{\frac{y^2}{4a}}^y \frac{x^2 - y^2}{x^2 + y^2} dx dy$. [5]
- Compute $\int \int_R \left(\frac{x+y}{2-x+y}\right)^4 dx dy$, where R is the square with vertices at $(1, 0)$, $(0, 1)$, $(-1, 0)$ and $(0, -1)$. [5]
- Evaluate $\int \int \int_V \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2}} dx dy dz$, where $V = \{(x, y, z) / \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1\}$. [5]
- Find the curvature and torsion of the curve $x = a \cos t$, $y = a \sin t$, $z = bt$. [5]
- Find the workdone in moving a particle once around a circle C in the xy -plane, if the circle has its centre at the origin and radius 2 units and force field is given by [5]

$$\vec{F} = (2x - y + 2z)\hat{i} + (x + y - z)\hat{j} + (3x - 2y - 5z)\hat{k}$$



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NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

Mid Semester Examinations, March-2025

Section: B

Course Name: Professional Communication

Date: 04/03/2025

Duration: 1.5 Hours

Course Code: HU150

Time: 10:00 am- 11:30 am

Max. Marks: 50

ANSWER ALL QUESTIONS

Read the Cases and Answer the Following Questions

4*5=20

Case Study 1:

You are a manager at a multinational company. Your team consists of members from diverse cultural and linguistic backgrounds. Recently, there has been a significant increase in miscommunication among team members, leading to delays in project timelines. Complaints have also been received about inadequate feedback during team meetings. Moreover, the grapevine communication within the team has led to the spread of false information about a policy change, causing unrest among employees. As a manager, you must address these challenges and improve the overall communication within the team.

1. Describe the types of noise evident in the given scenario. Propose two strategies to eliminate them effectively.
2. Explain the significance of proxemics and chronemics in a multicultural workplace like the one described in the case study. Provide examples.
3. Discuss how feedback can be made more effective in team meetings. Use the case study to illustrate your points.
4. Define the 7Cs of professional communication. Choose any two principles and explain how they can be applied to resolve the communication challenges in the case study.

Case Study 2:

5*5=25

During a project review meeting at an IT company, the team leader observed distinct listening styles among the team members. One member displayed active listening by asking clarifying questions and paraphrasing ideas. Another member exhibited superficial listening, appearing distracted and failing to grasp critical points. A third member used empathic listening to understand a colleague's concerns regarding workload. Some members leaned on evaluative listening, critically assessing the proposed solutions but neglecting the speaker's intent. Meanwhile, others practiced focused listening, fully

concentrating on the discussion and contributing effectively. The leader needs to train the team on improving their listening skills to enhance overall team performance.

5. Identify and differentiate the listening styles observed in the case study. Explain how each style affects communication within a team.
6. Discuss the importance of active and empathic listening in resolving team conflicts. Provide examples from the case study.
7. Propose strategies to help team members shift from superficial to focused listening. How can this improve team productivity?
8. Evaluate the role of evaluative listening in decision-making processes. When can it become a barrier to effective communication? Use examples to support your argument.
9. Suggest a training module to enhance listening skills in the team, incorporating the different listening styles mentioned in the case study.

Match the following

(5*1=05)

Column A	Column B
Olfactics	Maintaining distance in formal meetings
Chromatics	Perfumes or environmental scents.
Kinesics	Punctuality or time perceptions.
Proxemics	Body movements
Haptics	White color



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NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

Mid Semester Examinations, March-2025

Course Name: Engineering Chemistry/Chemistry

Date: 05.03.2025

Duration: 1.5 Hours

Course Code: CY 150

Time: 10-11.30 AM

Max. Marks: 50

ANSWER ALL QUESTIONS

1. Compare three major reasons affecting the T_g values of the polymers by giving the proper structure of the polymers and the corresponding T_g values 4 Marks
2. Explain a method that is suitable for commercial polymer synthesis. Explain with a proper diagram and examples of two polymers that can be synthesized by this method 3 Marks
3. Write the synthesis of any two conducting polymer 6 Marks
4. Write the classification of the polymers based on the intermolecular interaction. By giving each classification write an example of the synthesis of one polymer for each type 7 Marks
5. Classify the co-polymers, explain the classification, and give the proper example for each type 5 Marks
6. The sewage water collected from a sewage treatment plant is diluted to 17% for determination of COD. A 25 ml of diluted sewage water is refluxed with 25 ml of acidified $K_2Cr_2O_7$ in the presence of $HgSO_4$ and Ag_2SO_4 . The unreacted $K_2Cr_2O_7$ is titrated against 0.25 N FAS. Similarly, a blank titration is done with distilled water. During the experiment, 8.2 ml of FAS was consumed in the case of titration with sewage water and 19.5 ml of FAS was consumed in the case of blank titration. Calculate the COD in the units of PPM in the sewage water. 5 Marks
7. A litter borewell water was detected with 7.3 mg of $Mg(HCO_3)_2$ ($M_w = 146$), 16.2 mg of $Ca(HCO_3)_2$ ($M_w = 162$), 9.5 mg of $MgCl_2$ ($M_w = 95$), and 13.6 mg of $CaSO_4$ ($M_w = 136$). Calculate the temporary, permanent and total hardness in mg/L. 6 Marks
Further, convert the total hardness to the units of PPM, $^{\circ}Cl$ and $^{\circ}Fr$.
8. Describe reactions involved in corrosion due to Oxygen. Why does this reaction happen spontaneously? Further, describe how to overcome corrosion to oxygen. 6 Marks
9. What are zeolites? Describe the zeolite process of water treatment with an appropriate diagram and reactions. Also, list out the limitations of the zeolite process 5 Marks
10. Describe the electrodialysis method with an appropriate diagram. 3 Mark

All The Best



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National Institute of Technology Goa

Programme Name: B.Tech.

Mid Semester Examinations, March, 2025

Course Name: **Basics of Electronics Engineering**

Course Code: **EC150**

Date: 06/03/2025

Time: 10.00 AM - 11.30 AM

Duration: 1 Hour 30 Minutes

Max. Marks: 50

ANSWER ALL QUESTIONS

Instructions

- Answer all the questions. Assume suitable values if data is insufficient

Part A: 2 Marks Questions

1. Question 1: Plot input and output characteristic of BJT Common-Emitter (CE) configuration?
2. Question 2: In a BJT transistor, $\alpha = 0.998$, $I_B = 0.03$ mA. Find out collector current I_C and emitter current I_E .
3. Question 3: Plot I-V characteristic of PN junction diode. Discuss on knee voltage.
4. Question 4: What is a clipper. Show the circuit diagram.
5. Question 5: What are symbol representations of n-channel JFET, p-n-p BJT and n-p-n BJT.

Part B: 5 Marks Questions

1. What are the advantages of Field Effect Transistor (FET) over BJT ? Explain working principle of JFET and Draw characteristic.
2. An inverting op-amp circuit has an input resistor $R_{in} = 5k\Omega$ and a feedback resistor $R_f = 50k\Omega$. If the input voltage is $V_{in} = -0.2V$, determine:
 - (a) The voltage gain of the amplifier.
 - (b) The output voltage V_{out} .
 - (c) What is the maximum permissible input voltage?
3. Design the half-wave and full-wave rectifiers. Discuss the operation.

4. Answer the following about Bipolar Junction Transistors (BJTs):

- (a) Explain the physical operation of a BJT in the active region.
- (b) Differentiate between the cutoff, active, and saturation regions of a BJT.
- (c) Why is the base region of a BJT made very thin?
- (d) How does current amplification occur in a BJT?

Part C: 10 Marks Questions

1. (a) Discuss the construction of n-channel JFET and operation. What are the I-V characteristics.
- (b) The sinusoidal input with 6V p-p is given as input to the below circuit. Draw the input and output waveform. Assume the Zener Breakdown voltage for both diodes is 3 V.

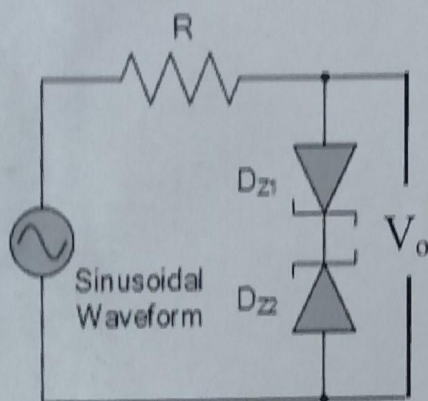
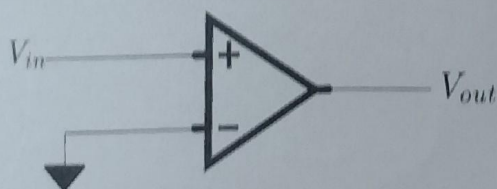
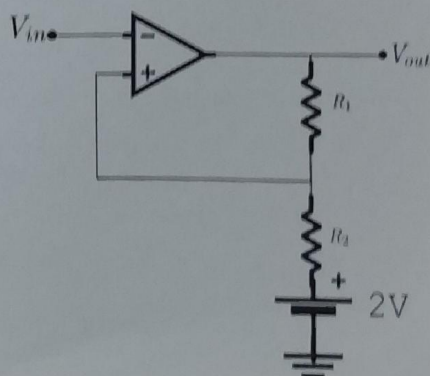


Figure 1

2. (a) An op-amp has gain of 1000 as in Figure 1. DC supply voltages are ± 12 V. Find out the output voltage for the input values (v_{in}) of (a) 10mV (b) 100mV
- (b) In Figure 2, $R_1 = R_2$, $V_{in} = 5V$, DC supply voltages are ± 12 V, what is the output voltage?



(a) Figure 1



(b) Figure 2.



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa - 403703

Programme Name: B.Tech.
Mid Semester Examinations, March-2025

Course Name: **Basics of Mechanical and Civil Engineering**

Date: 07-03-2025

Duration: 90 Minutes

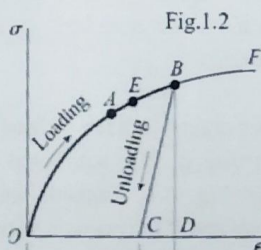
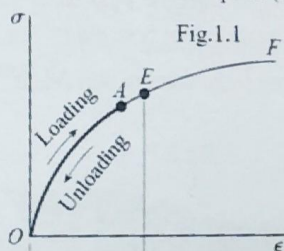
Course Code: **ME150**

Time: 10.00 AM to 11:30 AM

Max. Marks: 50

Instructions: Answer all the questions. Make an assumption and include it in the solutions if you believe any data is required.

1. (a) In the absence of any friction and other irreversibility, can a heat engine have an efficiency of 100 percent? Explain your answer. **(2 marks)**
- (b) Categorize the following properties into extensive and intensive: density, internal energy, square root of volume, melting point and boiling point, enthalpy, pressure, specific volume, surface tension, viscosity. The ratio of two extensive properties of the same object is (i) extensive or (ii) intensive property. Explain with an example. **(2 marks)**
- (c) Describe why, in the stress-strain diagram situations (point A and B) below, the unloading has taken a different path (Fig. 1.1 and 1.2). **(2 marks)**



- (d) Discuss the change in the volume in the below scenarios: **(2marks)**

- (i) cylindrical rod loaded with a normal force
(ii) square with a shear stress (given in Fig. 1.3)

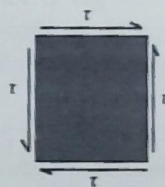
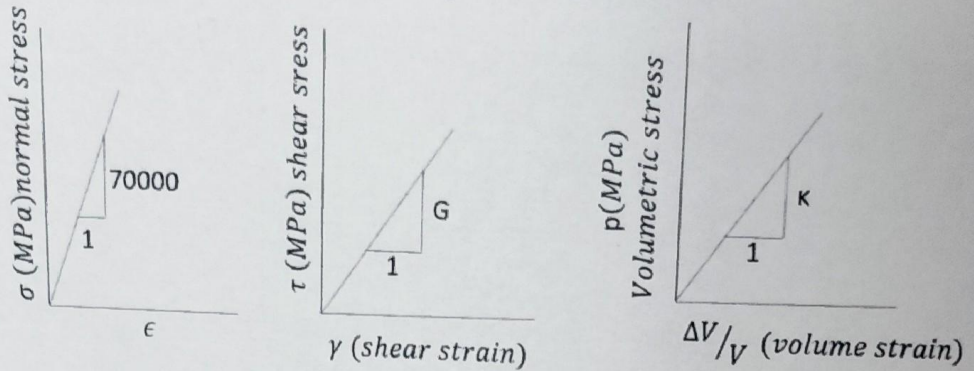


Fig.1.3

- (e) From a successful UTM tensile test, the slope of the Hookean region is calculated as given in the Fig. 1.4. Find the slopes G and κ for the other two plots given that poisson's ratio (ν) of the material is 0.25. **(2 marks)**

Fig.1.4



2. (a) Derived the steady flow energy equation for a single stream entering and a single stream leaving a control volume and explain the various terms it? **(5 marks)**

(b) A nozzle is a device for increasing the velocity of a steadily flowing stream. At the inlet to a certain nozzle, the enthalpy of the fluid passing is 3000 kJ/kg and the velocity is 60 m/s. At the discharge end, the enthalpy is 2762 kJ/kg. The nozzle is horizontal and there is negligible heat loss from it. (a) Find the velocity at exit from the nozzle. (b) If the inlet area is 0.1 m^2 and the specific volume at inlet is $0.187 \text{ m}^3/\text{kg}$, find the mass flow rate. (c) If the specific volume at the nozzle exit is $0.498 \text{ m}^3/\text{kg}$, find the exit area of the nozzle. **(5 marks).**

3. (a) A marine engineer claims to have devised an engine which can cruise on the sea based on schematics shown in Fig. 3.1. Explain, which statement has been violated if the claim is untrue. **(2.5 marks)**

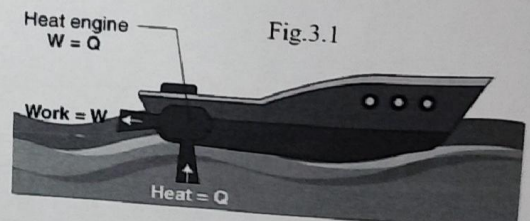


Fig.3.1

- (b) An engineer claims to have built a perfect refrigerator as shown in the Fig. 3.2. Explain, which statement has been violated if the claim is concocted. **(2.5 marks)**

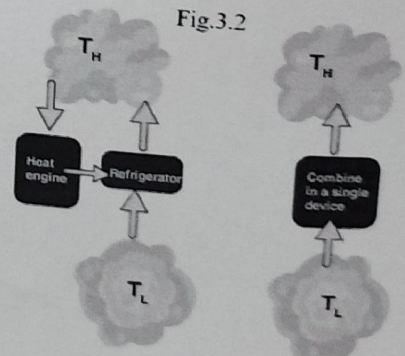


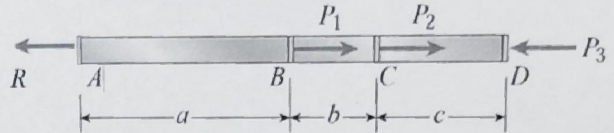
Fig.3.2

- (c) A gas of mass 1.5 kg undergoes a quasi-static expansion which follows a relationship $p = a + bV$, where a and b are constants. The initial and final pressures are 1000 kPa and 200 kPa respectively and the corresponding volumes are 0.20 m^3 and 1.20 m^3 . The specific internal energy of the gas is given by the relation $u = 1.5 pv - 85 \text{ kJ/kg}$, where p is the kPa and v is in m^3/kg . Calculate the net heat transfer and the maximum internal energy of the gas attained during expansion. **(5 marks)**

4. (a) Write short note on stress tensor (2 marks)

(b) Steel bar AD (see figure) has a cross-sectional area of $25 \times 10^{-5} \text{ m}^2$ and is loaded by forces $P_1=12\text{ kN}$, $P_2=8 \text{ kN}$ and $P_3=6 \text{ kN}$, $R=14 \text{ kN}$. The lengths of the segments of the bar are $a=1.5\text{m}$, $b=0.6\text{m}$, and $c=1\text{m}$. (5 marks)

(i) Assuming that the modulus of elasticity $E=200\text{ GPa}$, calculate the change in length δ of the bar. Does the bar elongate or shorten?

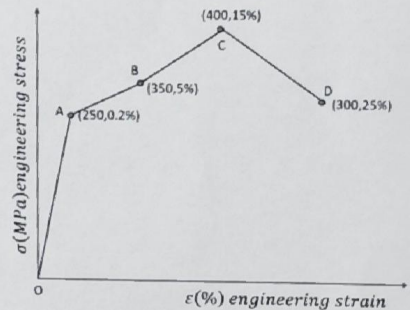


(ii) By what amount should the load P_3 be increased so that the bar does not change in length when the loads are applied?

(iii) If we would like to change the BC portion of the bar such that bar does not change in length when the loads are applied then find the change in the modulus of elasticity if the material has been changed.

(iv) If $P_2 = P_3$ which part of the bar will be completely useless.

(c) A hypothetical engineering stress-strain curve shown in the figure has four straight lines OA, AB, BC and CD. Find the true stress and true strain at points A, B, C and D plot it along with the engineering stress-strain curve. Explain the disparity between the engineering stress, engineering and true stress and true strain (closeness and separation at individual points). (3 marks)



5. Answer the following:

- What is the purpose of using National Building Code? (1 mark)
- Define the word "Orientation" in building Planning. (1 mark)
- What are the basic requirements of building? (2 marks)
- Briefly explain the components of residential building. (5 marks)
- List out the types of buildings as per National Building Code. (1 mark)



National Institute of Technology Goa

Kottamoll Plateau, Cuncolim, Salcete, South Goa, Goa - 403703

Programme Name: BTech

Mid-Semester, March 2025

Course Name : Health & Happiness (HH)

Course Code : HU151

Date : 7th Mar. 2025

Time Duration : 2:00 p.m. - 3:30 p.m.

Duration : 1 Hr. 30 Min.

Max. Marks : 50

- Answer all the questions neat & cleanly. **One (1) mark is considered for neat and clean Answer script.**
- No marks is there for irrelevant text/answer.
- **Any type of malpractice invites the cancellation of your answer script.**

1. Select the correct option (A, B, C, or D) and write your answer.

[1 × 10 = 10]

(a) What is the normal blood pressure for an adult?

- (A) 100/60 mmHg (B) 120/80 mmHg (C) 140/90 mmHg (D) 160/100 mmHg

(b) Where is the pulse commonly measured?

- (A) Neck and wrist (B) Elbow and knee (C) Ankle and chest (D) Shoulder and hip

(c) What is the function of the endocrine system?

- (A) Producing and eliminating hormones (B) Transporting oxygen
(C) Breaking down food (D) Fighting infections

(d) What happens when blood pressure is too high?

- (A) The heart stops beating (B) Blood circulation slows down
(C) The risk of heart disease increases (D) The body temperature rises significantly

(e) The majority of nutrient absorption occurs in which part of the digestive system?

- (A) Stomach (B) Small Intestine (C) Large Intestine (D) Esophagus

(f) Which of the following is NOT considered an essential nutrient?

- (A) Carbohydrates (B) Lipids (C) Fiber (D) Proteins

(g) What is the primary role of vitamin K in the human body?

- (A) Boosting the immune system (B) Aiding in calcium absorption
(C) Facilitating blood clotting (D) Acting as a coenzyme in energy metabolism

(h) Which class of phytochemicals is responsible for the red, blue, and purple colors in fruits and vegetables?

- A) Carotenoids (B) Flavonoids (C) Anthocyanins (D) Isoflavones

(i) What is the normal respiration rate for an adult at rest?

- (A) 8-12 breaths per minute (B) 12-16 breaths per minute
(C) 16-24 breaths per minute (D) 24-30 breaths per minute

(j) What is the primary function of the respiratory system?

- (A) Transport nutrients (B) Regulate body temperature
(C) Exchange gases (oxygen and carbon dioxide) (D) Control muscle movement

2. Answer the following questions.

[2 × 8 = 16]

- (a) How many dhatus are present in the body? Name them.
- (b) What is Vital Parameter? Name the Vital Parameters.
- (c) What are GI Motility disorders?
- (d) What is Prakriti? How many Prakriti are there? Name them.
- (e) What is the meaning of Gandusha?
- (f) What are the benefits of phytochemicals?
- (g) What are the importance of cinnamons?
- (h) What types of circadian rhythm disorders are found?

3. Write a short note on each of the following topics.

[4 × 4 = 16]

- (a) Abhyanga
- (b) Characteristics of Kapha Prakriti
- (c) Components of Food
- (d) Vitamin B and its Importance

4. Explain the 12 postures of Surya Namaskar. Name them. Explain what are the advantages of Surya Namaskar. [7]

----- All The Best -----